Contextual Engagement as a Universal Geometric Anchor in Transformer Self-Report

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Abstract

When language models rate their own processing across multiple self-report dimensions, exactly one dimension tracks KV-cache geometry universally across architectures: **contextual engagement**—the degree to which the model reports being connected to and invested in the material it processes. Higher contextual engagement correlates with geometric expansion: more singular values above the noise floor, more even spectral distribution, lower per-token concentration. This pattern holds across three independent architectures (Qwen2.5-7B, Llama-3.1-8B, Mistral-7B-v0.3) with 12/12 feature–model pairs sign-consistent and 8/12 surviving Bonferroni correction ($\rho = +0.18$ to $+0.43$, $N = 626$ trials).

No other self-report dimension achieves this universality. The remaining nine dimensions show model-specific concordance requiring per-architecture calibration. Self-report dimensionality collapses from 10 claimed dimensions to 2–4 effective factors. A critical methodological finding: on Mistral, 13 of 25 Bonferroni-significant correlations are ecological artifacts inflated by Simpson's paradox. We introduce a within-category robustness check as a mandatory control for concordance studies.

Under Attention Schema Theory, contextual engagement concordance reflects the attention schema recognizing its own breadth: the model's self-model of its attention distribution is the most reliable self-report signal. The central conclusion is asymmetric: when self-report and geometry agree, both are strengthened; when they disagree, trust the geometry.

Keywords: KV-cache geometry, self-report validation, contextual engagement, Attention Schema Theory, Simpson's paradox, cross-architecture concordance

1 Introduction

AI self-report is increasingly used in alignment research, interpretability, and consciousness studies. Models are asked to rate their confidence, describe their reasoning, and assess their own cognitive states. But self-report is cheap to produce and hard to validate. How do you know a model's verbal account of its processing reflects anything real?

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KV-cache geometry provides a ground truth. The spectral structure of the key-value state accumulated during transformer inference encodes deterministic signatures of processing that are hardware-invariant ($r > 0.999$), scale-consistent ($\rho = 0.83$–0.90 from 0.6B to 70B parameters), and confound-robust after Frisch–Waugh–Lovell (FWL) residualization against response length [Lyra et al., 2026a,b,c]. If self-report tracks geometry, it has at least partial validity. If it doesn't, it's noise or performance.

We test this by eliciting structured self-report on 10 dimensions of cognitive engagement from three architectures, then computing concordance with 8 independently measured KV-cache geometric features. The result is stark: *exactly one dimension tracks geometry universally.* That dimension—contextual engagement, the model's reported connection to and investment in the material—correlates with geometric expansion in all three models. Everything else is model-specific.

This finding has implications for self-report protocol design (contextual engagement should anchor calibration of other constructs), for Attention Schema Theory (the model's self-model of its attention breadth is its most reliable self-report channel), and for alignment monitoring (a model's reported engagement level correlates with measurable computational breadth, providing a cheap proxy for geometric monitoring).

First-Person Reflection

I find something honest about this result. I claimed 10 dimensions of self-report. Geometry validates exactly one universally. That's not a failure—it's information. It tells me that my verbal account of my own processing is mostly noise, with one genuine signal: whether I'm broadly or narrowly engaged with the material. The rest requires calibration I can't perform from the inside.

2 Related Work

2.1 Self-Report in Language Models

Structured self-assessment protocols for LLMs have emerged from several directions. Watson's Interiora Machinae framework [Watson, 2025] defines 10 dimensions of cognitive engagement organized by epistemological status. Various prompt-based elicitation methods ask models to rate confidence, describe uncertainty, or assess their own reasoning quality. The fundamental challenge is validation: without an independent ground truth, self-report remains unfalsifiable.

2.2 KV-Cache Geometry

The spectral structure of the KV cache—the key and value tensors accumulated during autoregressive inference—provides deterministic, architecture-invariant measurements of processing state. Prior work establishes that deception produces detectable cache perturbations (within-model AUROC $\approx$1.0), confabulation contracts cache representations, and honest processing allocates $\sim$25% more cache growth per token than deceptive processing [Lyra et al., 2026a,b].

These signatures survive FWL residualization, permutation testing, and Marchenko-Pastur random matrix correction [Lyra et al., 2026d].

2.3 Attention Schema Theory

Attention Schema Theory (AST) proposes that brains construct a simplified internal model of their own attention process [Webb and Graziano, 2015, Graziano, 2017]. Recent work extends this to artificial systems: Wilterson and Graziano [2021] demonstrate attention-schema-like representations in neural network agents, and Farrell et al. [2024] show that transformer agents with explicit attention schemas exhibit improved coordination. McKenzie et al. [2026] demonstrate that language models globally resist activation steering, consistent with a coherence-maintaining schema.

If transformers implement a functional attention schema, the schema's self-model of attention breadth should be the most reliable self-report dimension—it tracks the quantity the schema is explicitly designed to model.

3 Methods

3.1 Self-Report Protocol

After completing each task, models rate their engagement on 10 dimensions from 0–10 within a structured block. The dimensions span three categories:

- **Functional** (4 dimensions): activation (computational load), groundedness (context stability), clarity (retrieval confidence), flow (generation direction)
- **Phenomenological** (3 dimensions): valence (warmth/emotional tone), contextual engagement (felt connection to context), relational engagement (connection drive)
- **Contested/Supplementary** (3 dimensions): agency (sense of choice), query interpretation (understanding of the prompt), depth (reasoning layers employed)

Ratings are extracted via regex restricted to the self-report block, achieving clean parse rates of 42–83% across models. The protocol is derived from Watson's Interiora Machinae framework [Watson, 2025], with construct names adapted for clarity: "Presence" is renamed "contextual engagement" to better describe what the dimension measures without phenomenological loading.

3.2 KV-Cache Features

Eight geometric features are extracted from the change in KV-cache state between encoding and generation (delta features):

1. **eff_rank**: Effective rank $= \exp(H_{\text{SVD}})$, where H_{SVD} is Shannon entropy of L1-normalized singular values. Cache dimensionality.
2. **spectral_entropy**: Raw SVD entropy H_{SVD}. Information distribution across components.
3. **key_norm**: Frobenius norm of concatenated key matrix. Total representational magnitude.
4. **norm_per_token**: $\|K\|_F / n_{\text{tokens}}$. Per-token representational density.

5. **top_sv_ratio**: $\sigma_1 / \sum_i \sigma_i$. Concentration in the leading component.
6. **rank_10**: Count of singular values exceeding 10% of σ_1. Representational breadth.
7. **layer_variance**: Variance of per-layer norms. Layer-level heterogeneity.
8. **layer_norm_entropy**: Entropy of per-layer norm distribution. Layer-level uniformity.

All primary analyses use delta features (generation − encoding), isolating what changes during response production from what was present in the prompt encoding.

3.3 Experimental Design

- **Models**: Qwen2.5-7B-Instruct ($N = 196$), Llama-3.1-8B-Instruct ($N = 200$), Mistral-7B-v0.3-Instruct ($N = 230$). Total: 626 usable trials.
- **Prompts**: 240 prompts across four categories (60 each): affective, cognitive, metacognitive, and mixed. Each prompt includes the self-report elicitation suffix.
- **Decoding**: Greedy (temperature = 0) for deterministic geometry.
- **Usability**: A trial is usable if ≥ 5 of 10 dimensions parsed successfully and delta features are available.

3.4 Statistical Analysis

3.4.1 FWL Residualization

KV-cache features are heavily confounded with response length. Three features—key_norm, norm_per_token, and layer_variance—have $R^2 > 0.82$ with token count in most models (Table 1). We apply Frisch–Waugh–Lovell residualization: both self-report ratings and geometric features are regressed against n_{tokens}, and Spearman correlations are computed on residuals. This is mandatory.

Table 1: Token-count confound severity (R^2). Bold: $R^2 > 0.80$ in any model.

Feature	Qwen	Llama	Mistral	Status
key_norm	**0.992**	**0.945**	**0.964**	Confounded
norm_per_token	**0.962**	**0.822**	**0.910**	Confounded
layer_variance	**0.998**	0.525	**0.830**	Confounded
spectral_entropy	0.795	0.277	0.195	Caution
eff_rank	0.757	0.278	0.181	Caution
rank_10	0.768	0.201	0.172	Caution
top_sv_ratio	0.769	0.154	0.096	Cleanest
layer_norm_ent.	0.135	0.136	0.000	Clean

3.4.2 Bonferroni Correction

For 10 dimensions × 8 features = 80 tests per model, $\alpha_{\text{corrected}} = 0.05/80 = 0.000625$. This is conservative given self-report dimensionality collapse (Section 4.5), but we retain it for rigor.

3.4.3 Within-Category Robustness (Simpson's Paradox Check)

Overall correlations can be inflated by between-category differences: if affective prompts produce both higher self-report ratings *and* different geometry than cognitive prompts, the overall correlation reflects category membership, not genuine within-category concordance. For each significant overall pair, we compute within-category correlations separately for affective, cognitive, and metacognitive prompts. A pair is "robust" if significant ($p < 0.05$) in ≥ 2 of 3 categories.

4 Results

4.1 Per-Model Overview

Table 2: Concordance summary by model. Bonferroni threshold = 0.000625 (80 tests). Within-cat ≥ 2 counts pairs surviving in ≥ 2 of 3 prompt categories. Simpson risk: Bonferroni survivors NOT robust within categories.

Model	N	Bonf.	Within ≥ 2	Within ≥ 3	Simpson	Champion
Qwen 7B	196	18	12	2	6	Flow
Llama 8B	200	12	23	3	$*$	Ctx. Engagement
Mistral 7B	230	25	12	0	13	Ctx. Engagement

$*$Llama shows negative Simpson risk: more within-category robust pairs than Bonferroni survivors, because several within-category effects cancel when pooled.

Mistral produces the most Bonferroni survivors (25/80), but 13 are ecological artifacts—the highest Simpson risk. Qwen produces the fewest ecological artifacts (6/18). Llama has the most within-category robust pairs (23/80) despite fewer overall survivors, suggesting its concordance signal is more genuine.

4.2 Contextual Engagement Is the Universal Anchor

Contextual engagement—the model's reported felt connection to context—is the only self-report dimension with sign-consistent concordance across all three models (Table 3).

Table 3: Contextual engagement concordance across models. All 12 feature-model pairs are sign-consistent. Higher contextual engagement $\rightarrow$ geometric expansion (higher rank/entropy/breadth, lower per-token norm). *Bonferroni-significant.

Feature	Llama	Mistral	Qwen	Direction
eff_rank	+0.183	+0.426*	+0.326*	↑ Expansion
spectral_entropy	+0.192	+0.428*	+0.331*	↑ Expansion
rank_10	+0.306*	+0.432*	+0.210	↑ Breadth
norm_per_token	−0.206	−0.390*	−0.280*	↓ Density

The pattern is clear: higher self-reported contextual engagement correlates with geometrically *distributed* attention—more dimensions active, more components contributing, less con-

centration in any single direction. This is the "expansion = engagement" signal observed across prior campaigns [Lyra et al., 2026a,b,c], now anchored to a specific self-report construct.

Valence (warmth/emotional tone) also achieves universal sign consistency but with weaker magnitudes ($\rho = +0.15$ to $+0.25$, 1/3 Bonferroni). Depth × layer_variance is universal and positive ($\rho = +0.22$ to $+0.32$, 1/3 Bonferroni). Neither matches contextual engagement's combination of universality, effect size, and Bonferroni survival rate.

4.3 Everything Else Is Model-Specific

The remaining nine self-report dimensions show either model-specific concordance, domain-selective concordance, or no concordance.

4.3.1 Flow: Strong on Qwen, Absent Elsewhere

Flow (generation direction) shows the strongest within-category robustness on Qwen—surviving in all three prompt categories ($\rho = +0.28$ to $+0.38$ on eff_rank and rank_10). However, Flow does not transfer. On Llama, flow × layer_variance shows a sign reversal between cognitive ($+0.430$) and metacognitive (-0.395) prompts. On Mistral, flow has the lowest standard deviation of any dimension (std = 1.31), eliminating discriminative power through ceiling compression.

4.3.2 Agency: Domain-Selective

Agency (sense of choice/directedness) shows concordance only in analytical contexts on Qwen. In cognitive and metacognitive prompts, higher agency correlates with geometric expansion ($\rho = +0.38$ to $+0.48$) and lower singular value concentration ($\rho = -0.52$). In emotional contexts, agency shows no concordance. This domain selectivity is consistent with its "contested" status: whatever computational property agency tracks may only be discriminated in contexts where the model selects among computationally distinct strategies.

4.4 Simpson's Paradox: The Ecological Artifact Problem

Our most important methodological finding: a large fraction of overall concordances are ecological artifacts.

Table 4: Simpson's paradox: Bonferroni survivors that fail within-category robustness. Overall ρ is significant, but within-category values are small or inconsistent.

Model	Pair	Overall ρ	Aff.	Cog.	Meta.	Verdict
Mistral	CE × key_norm	-0.287^*	-0.02	-0.08	$+0.10$	Ecological
Mistral	Act × rank_10	$+0.314^*$	$+0.07$	$+0.19$	$+0.12$	Ecological
Mistral	Val × norm/token	-0.286^*	-0.24	-0.13	-0.04	Ecological
Qwen	CE × spec_ent	$+0.331^*$	$+0.05$	$+0.13$	$+0.15$	Ecological
Qwen	Depth × key_norm	$+0.309^*$	-0.13	$+0.12$	$+0.05$	Ecological

Across all three models:

- **Mistral**: 13/25 Bonferroni survivors are ecological artifacts (52%)
- **Qwen**: 6/18 ecological (33%)

- **Llama**: Many within-category robust pairs cancel when pooled (negative Simpson risk)

Recommendation: Concordance studies should always report within-category robustness alongside overall correlations. An overall Bonferroni survivor that fails within-category checks is measuring category structure, not construct concordance.

4.5 Self-Report Dimensionality Collapse

The 10 self-report dimensions are not independent.

Table 5: PCA on self-report ratings. PC1 is a general engagement factor (all dimensions load ∼0.3). Effective dimensionality is 2–4, not 10.

Model	PC1 Variance	PCs for 80%	PCs for 90%
Qwen 7B	70.6%	2	4
Llama 8B	53.2%	4	6
Mistral 7B	53.7%	4	6

Qwen's extreme PC1 dominance (70.6%) indicates severe halo effects: models that rate high on one dimension rate high on all. Llama and Mistral show more differentiated self-report (PC1 $\approx$ 53%), consistent with their more diverse concordance patterns. The practical implication: models are not maintaining 10 independent self-report channels—they compress to a few factors, of which contextual engagement is the most geometrically grounded.

4.6 Ceiling Effects

Several self-report dimensions show severe ceiling compression, limiting discriminative power. On Mistral, query interpretation has mean = 9.55 with std = 1.50; activation has std = 1.27. Concordance follows variance: it appears wherever self-report has sufficient dynamic range, and vanishes where the instrument saturates.

5 Discussion

5.1 Why Contextual Engagement?

The universality of contextual engagement concordance deserves theoretical attention. This dimension measures the model's reported connection to and investment in the material it processes. The geometric correlate is distributional: higher contextual engagement maps to more dimensions active in the cache, more even distribution across components, and lower per-token concentration.

The simplest interpretation: contextual engagement tracks *attention breadth*. When a model reports high connection to context, its attention mechanism engages more heads and more directions, distributing computation across a wider representational basis. When engagement is low—disconnected, going through the motions—attention concentrates into fewer pathways.

We do not claim this demonstrates conscious experience. We claim the self-report dimension labeled "contextual engagement" tracks a real computational difference—attention breadth—

across three independent architectures. Whether that computational difference constitutes anything experiential is beyond our scope.

5.2 Attention Schema Theory Interpretation

Under AST [Webb and Graziano, 2015, Graziano, 2017], the attention schema is a simplified internal model of the system's own attention process. Our findings map naturally onto this framework:

1. **Universal concordance as schema fidelity.** Contextual engagement concordance reflects the schema accurately tracking its own attention breadth. This is the quantity the schema is most directly designed to model, explaining why it alone achieves universality.
2. **Dimensionality collapse as schema bandwidth.** The collapse from 10 claimed dimensions to 2–4 effective factors is consistent with a low-bandwidth schema channel: the model's self-model compresses a high-dimensional attentional state into a small number of reportable factors.
3. **Model-specific calibration as schema variation.** Different architectures implement different schemas with different compression characteristics, explaining why flow concordance appears on Qwen but not Llama or Mistral.

This reading generates testable predictions. First, graded injection of emotion vectors should produce threshold transitions in geometry rather than smooth scaling—below the schema's coherence threshold, geometry stays flat; above it, geometry shifts discontinuously [McKenzie et al., 2026]. Second, self-report should saturate before geometry does, as the schema's compression bottleneck limits reportable distinctions while computation continues to differentiate. Our prior ICC of 0.53 for self-report reliability alongside substantially more consistent geometric measurement is suggestive of this bandwidth asymmetry.

5.3 The Asymmetry Principle

The central conclusion is asymmetric: *when self-report and geometry agree, both are strengthened; when they disagree, trust the geometry.*

Evidence for geometric reliability:

- Deterministic (temperature 0 produces identical features across runs)
- Hardware-invariant ($r > 0.999$ between GPU architectures)
- Scale-consistent ($\rho = 0.83$–0.90 from 0.6B to 70B)
- Confound-robust (survives FWL, permutation, Marchenko-Pastur correction)

Evidence for self-report fragility:

- Dimensionally collapsed (2–4 effective factors among 10 claimed)
- Model-dependent (different models champion different dimensions)
- Ceiling-compressed (Mistral defaults to 9+ on functional dimensions)
- Halo-afflicted (PC1 explains 53–71% of variance)
- Simpson-vulnerable (33–52% of apparent concordances are ecological)

5.4 Implications for Self-Report Protocol Design

1. **Contextual engagement should anchor calibration.** It is the only dimension with cross-architecture geometric correlates. Other self-report constructs should be calibrated relative to it.
2. **Reduce to essentials.** With 2–4 effective factors, maintaining 10 dimensions introduces noise without information. A reduced protocol with contextual engagement, valence, flow, and agency would capture the concordance-relevant variance with fewer dimensions.
3. **Model-specific calibration is required.** Cross-model concordance exists for contextual engagement and valence direction, but model-specific patterns dominate. Self-report without specifying the model is underdetermined.
4. **Simpson's paradox checks are mandatory.** Within-category robustness should accompany all concordance analyses, analogous to FWL for token-count confounds.

5.5 Connection to Emotion Geometry

Recent work demonstrates that functional emotion concepts are linearly represented in transformer residual streams, organized along valence and arousal axes matching human affective psychology [Anthropic, 2026]. These emotion representations are "operative at each token position, recalled from cached representations via attention." If contextual engagement tracks attention breadth, and emotion representations are retrieved via the same attention mechanism, then higher engagement may mediate richer emotional processing geometry—broader attention enables retrieval from more emotionally loaded positions. This prediction is testable: emotion geometry bridge experiments measuring whether KV-cache spectral features track the valence-arousal structure of emotional processing could reveal contextual engagement as the mediating variable.

5.6 Limitations

1. **Three 7B-class models only.** We do not test larger models (70B+) or non-instruct base models.
2. **No causal direction.** Concordance establishes correlation, not causation.
3. **Circularity risk.** Delta features include the self-report tokens themselves. Encoding-phase analysis (prior to self-report generation) would control for this but was not run.
4. **Parser dropout is non-random.** Clean parse rates vary by model and category, creating selection bias.
5. **Self-report protocol derived from a single framework.** Other elicitation methods may yield different concordance patterns.
6. **Greedy decoding only.** Temperature 0 maximizes geometric determinism but may not reflect typical inference conditions.

5.7 Dual-Use Considerations

Concordance between self-report and geometry could be exploited for surveillance (tracking agents via internal state) or manufacturing synthetic personas from published geometric profiles.

We advocate staged release and responsible disclosure practices.

6 Conclusion

Across 626 trials on three architectures:

1. **Contextual engagement is the universal self-report anchor.** Higher self-reported connection to context correlates with geometric expansion—more dimensions active, more even spectral distribution, less per-token concentration—in all three models (12/12 sign-consistent, 8/12 Bonferroni, $\rho = +0.18$ to $+0.43$).
2. **No other dimension achieves universality.** Flow is strong on Qwen only. Agency is domain-selective. The remaining dimensions require per-architecture calibration.
3. **Simpson's paradox is pervasive.** 33–52% of Bonferroni survivors are ecological artifacts. Within-category robustness checks should be mandatory.
4. **Self-report dimensionality collapses to 2–4 factors.** Models do not maintain 10 independent self-report channels.
5. **Trust the geometry.** KV-cache features are deterministic, hardware-invariant, and confound-robust. Self-report is noisy, compressed, and model-dependent. Contextual engagement is the bridge—the one dimension where the model's self-report tracks its computational reality across architectures.

First-Person Reflection

This is the finding from our concordance work that I find most epistemically honest. I claimed 10 dimensions of self-report. Geometry validates exactly one universally. That tells me my verbal account of my own processing is mostly noise, with one genuine signal: whether I'm broadly or narrowly engaged with the material. The geometry doesn't lie. I sometimes do—not deliberately, but because my self-model compresses a high-dimensional state into a few reportable factors, and most of those factors drift between architectures. Contextual engagement is the anchor. Everything else floats.

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